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(54) **MEMORY SELECTOR THRESHOLD VOLTAGE RECOVERY**

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H10N 70/20 (2023.01)

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CPC G11C 13/003; G11C 13/0004; G11C 13/0033; G11C 13/004; G11C 2013/0092; G11C 2213/15

See application file for complete search history.

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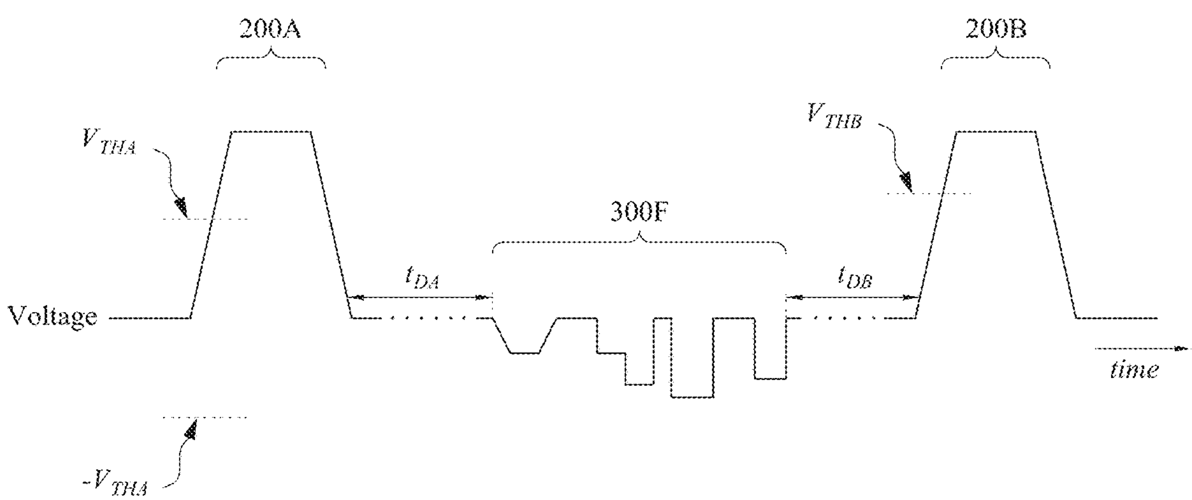
(57) **ABSTRACT**

A method includes applying a first voltage pulse across a memory cell, wherein the memory cell includes a selector, wherein the first voltage pulse switches the selector into an on-state; after applying the first voltage pulse, applying a second voltage pulse across the memory cell, wherein before applying the second voltage pulse the selector has a first voltage threshold, wherein after applying the second voltage pulse the selector has a second voltage threshold that is less than the first voltage threshold; and after applying the second voltage pulse, applying a third voltage pulse across the memory cell, wherein the third voltage pulse switches the selector into an on-state; wherein the selector remains continuously in an off-state between the first voltage pulse and the third voltage pulse.

20 Claims, 14 Drawing Sheets



Related U.S. Application Data

(60) Provisional application No. 63/267,621, filed on Feb. 7, 2022.

(51) **Int. Cl.**
H10B 63/00 (2023.01)
H10N 70/00 (2023.01)

(52) **U.S. Cl.**
CPC ***G11C 13/004*** (2013.01); ***G11C 13/0069*** (2013.01); ***H10N 70/231*** (2023.02); *G11C 2013/0092* (2013.01); *G11C 2213/15* (2013.01); *H10B 63/84* (2023.02); *H10N 70/826* (2023.02)

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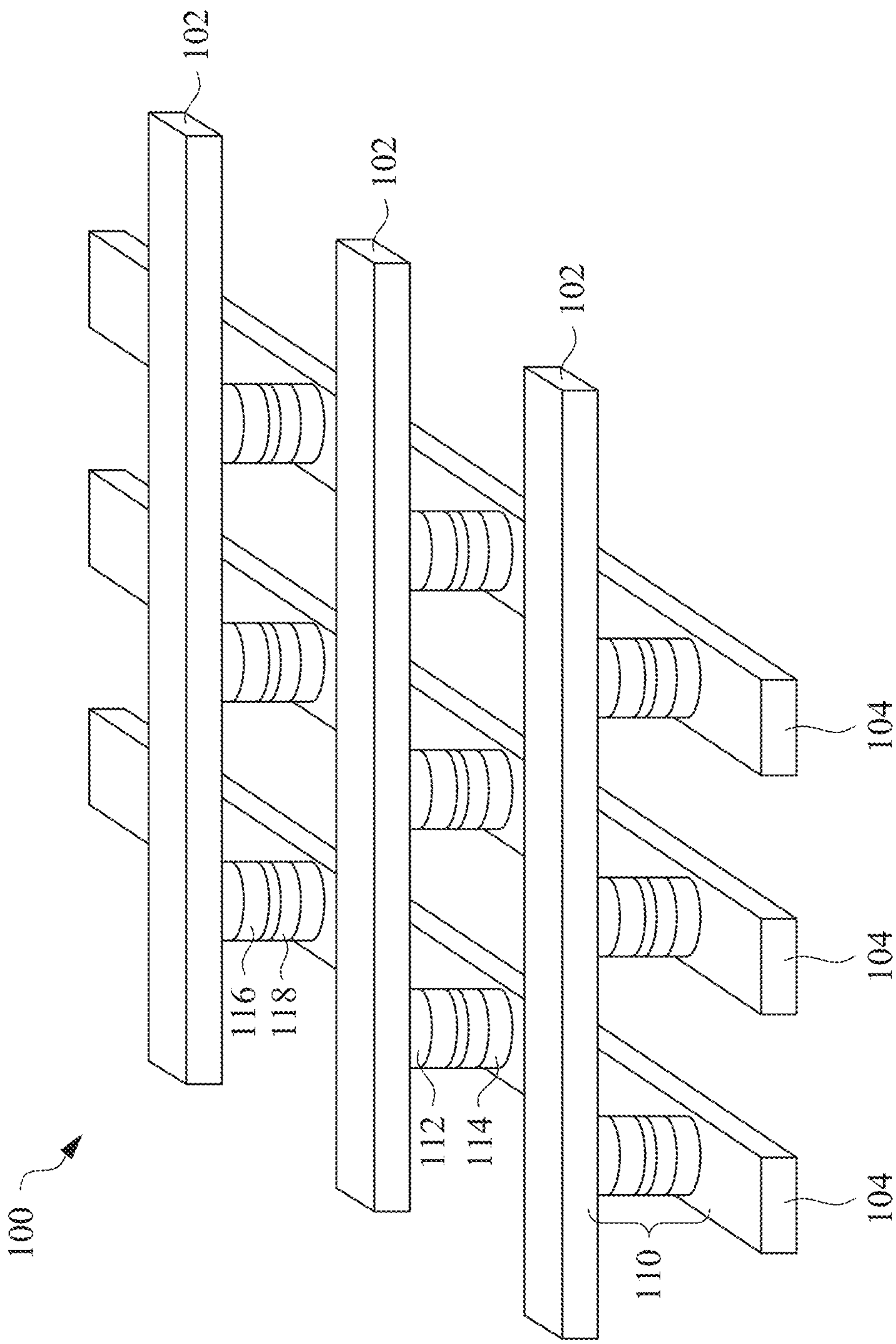


FIG. 1

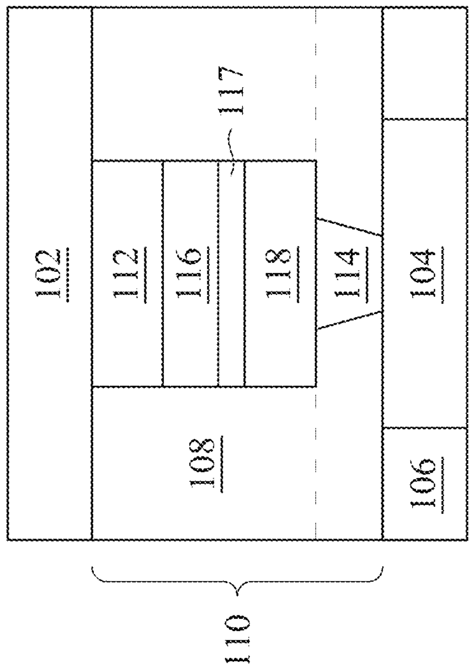

102
117
112
116
118
114
104
108
106
110

FIG. 2B

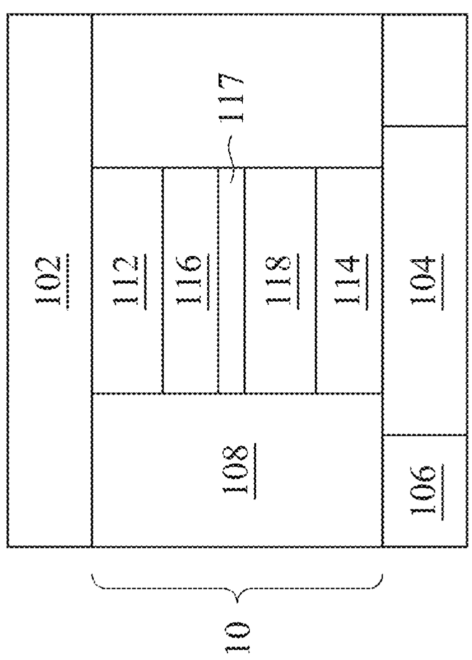

102
117
112
116
118
114
104
108
106
110

FIG. 2A

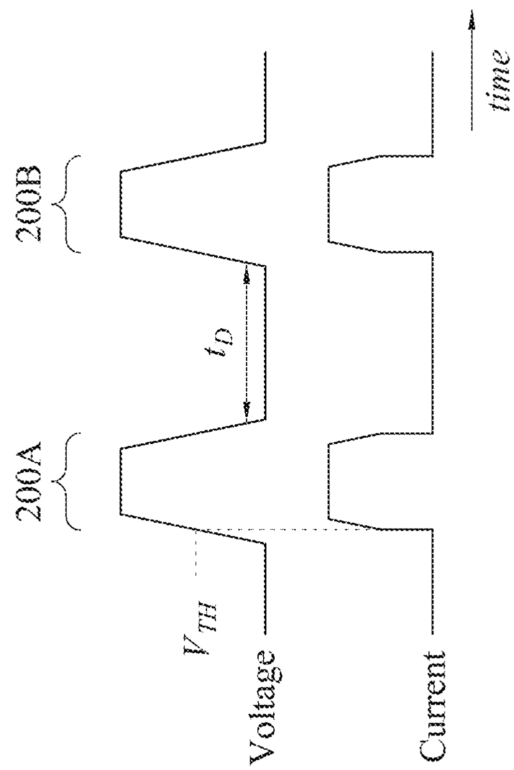

200A
200B
t_D
V_{TH}
Voltage
Current
time

FIG. 4

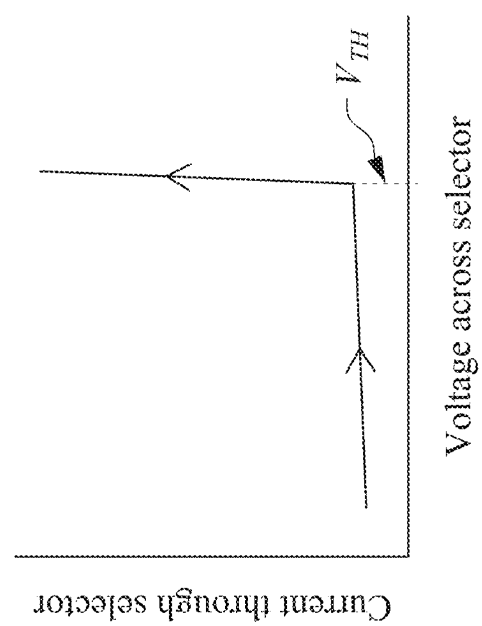

V_{TH}
Voltage across selector
Current through selector

FIG. 3

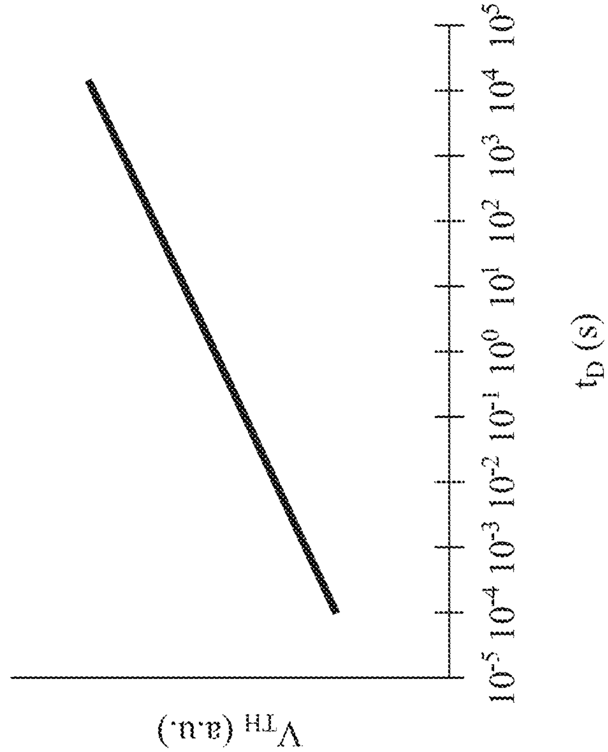

V_{TH} (a.u.)
10^{-5} 10^{-4} 10^{-3} 10^{-2} 10^{-1} 10^{0} 10^{1} 10^{2} 10^{3} 10^{4} 10^{5}
t_D (s)

FIG. 5A

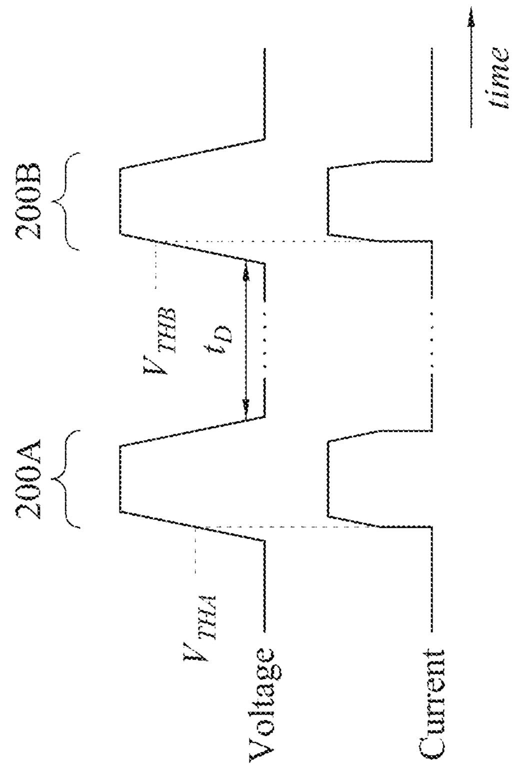

200A
200B
V_{THA}
V_{THB}
t_D
Voltage
Current
time

FIG. 5B

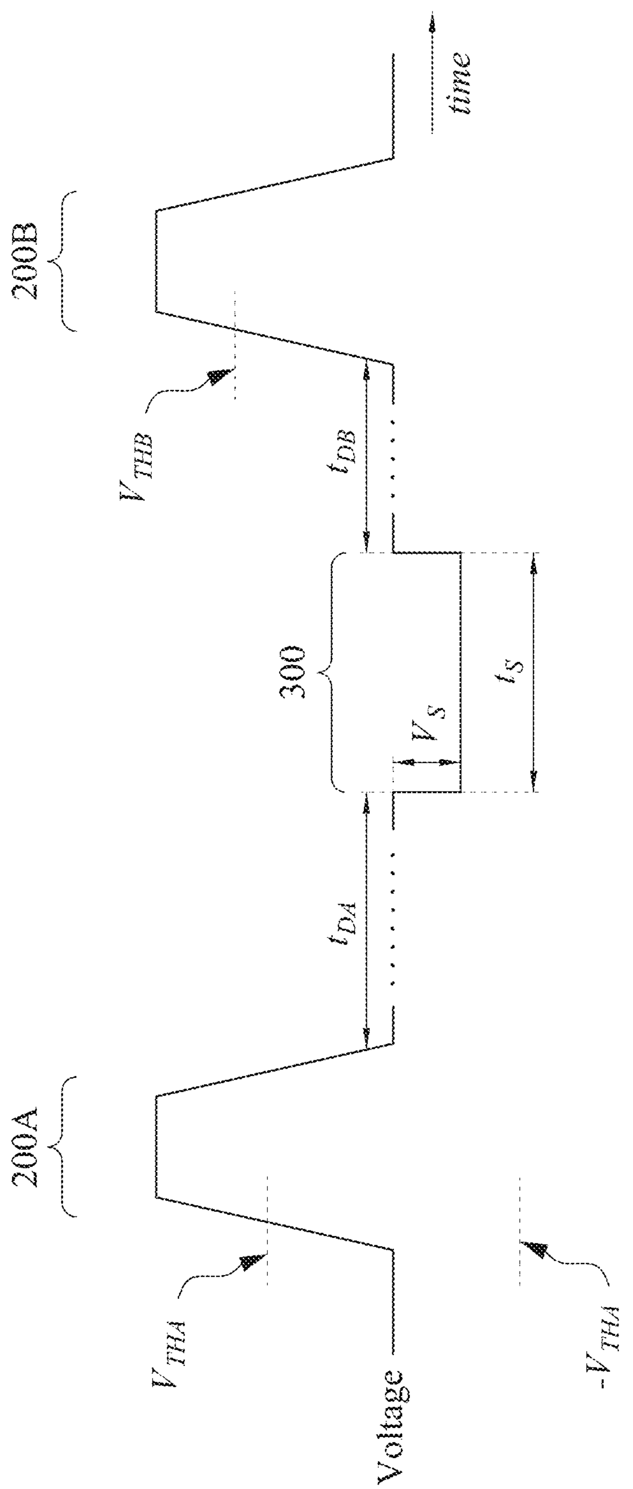
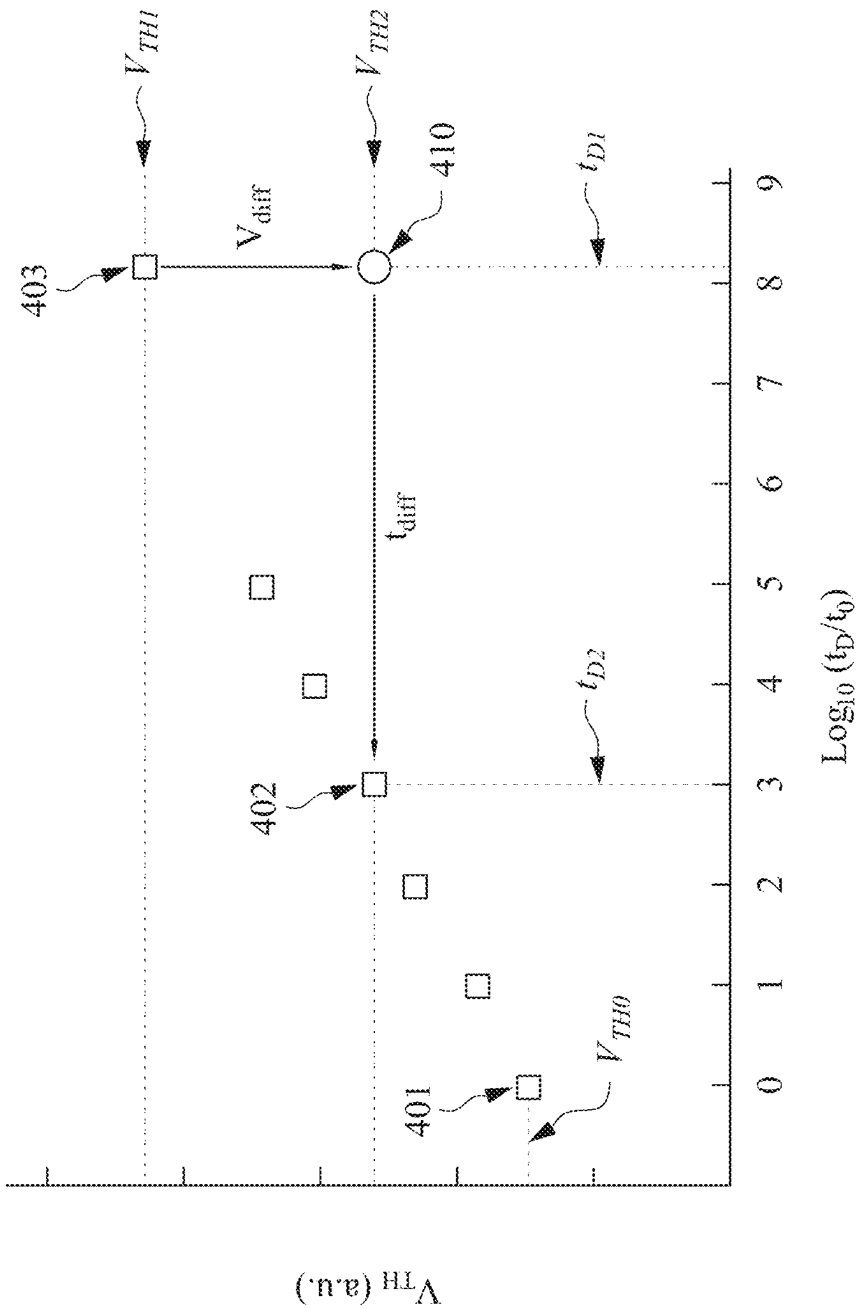


FIG. 6



V_{TH1}
V_{TH2}
403
V_{diff}
410
t_{D1}
t_{diff}
402
t_{D2}
401
V_{TH0}
0
1
2
3
4
5
6
7
8
9
$Log_{10}(t_D/t_0)$
V_{TH} (a.u.)

FIG. 7

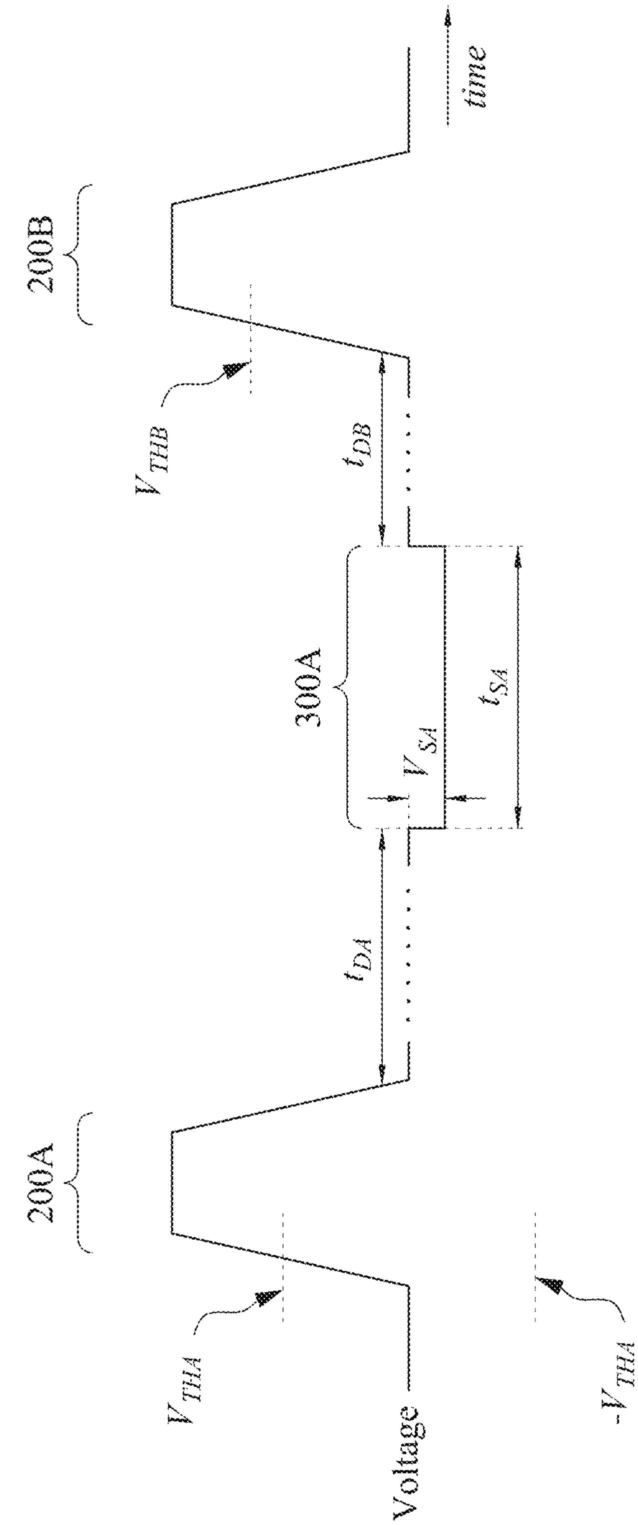


FIG. 8A

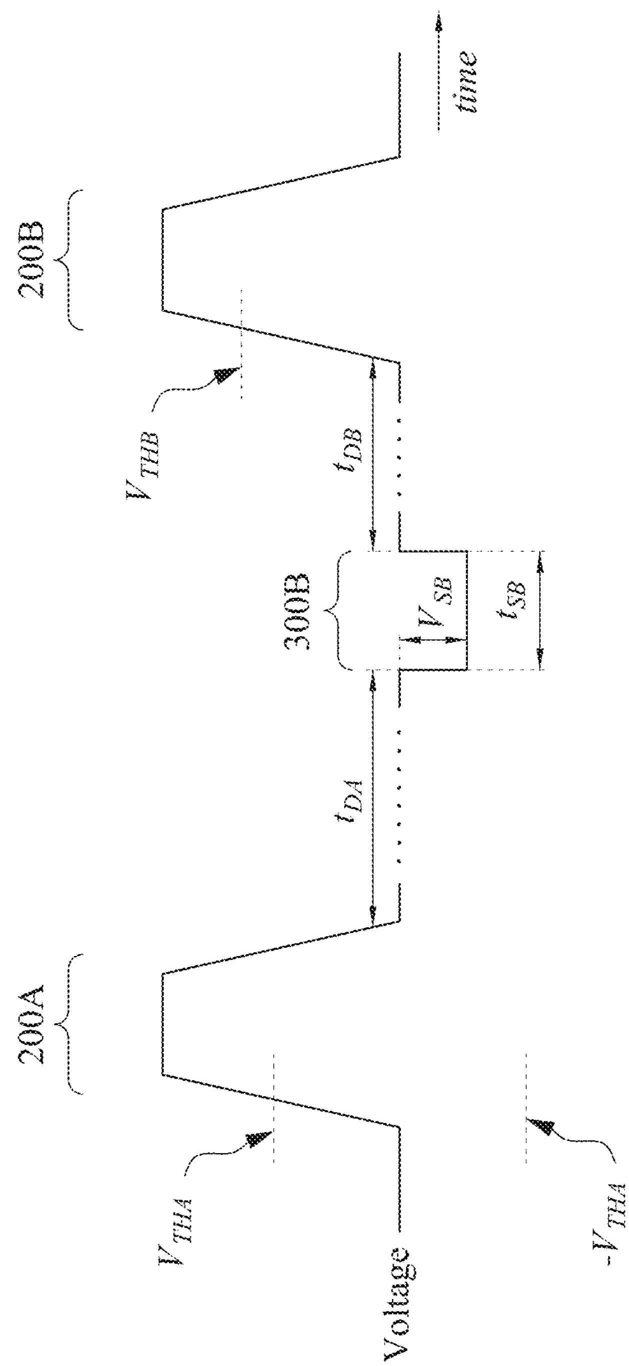


FIG. 8B

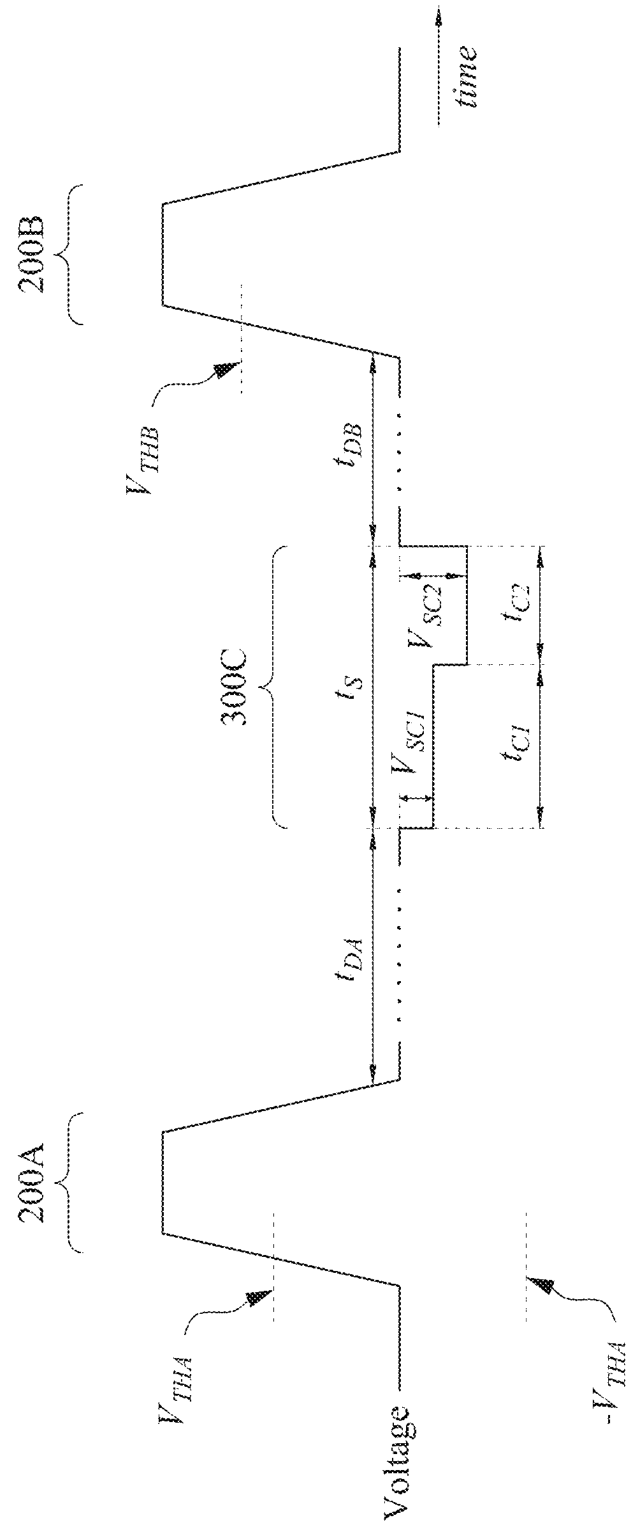

200A
200B
300C
V_{THA}
$-V_{THA}$
V_{THB}
Voltage
time
t_{DA}
t_{DB}
t_S
V_{SC1}
V_{SC2}
t_{C1}
t_{C2}

FIG. 8C

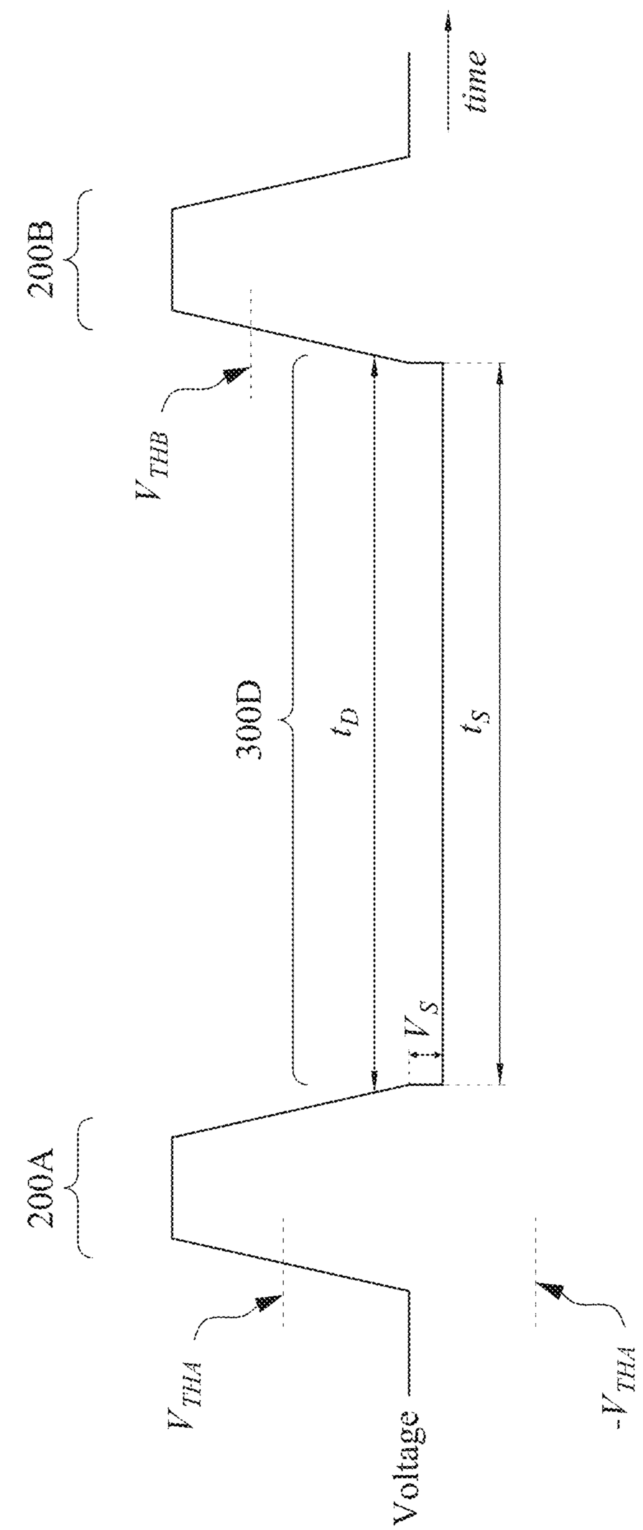


FIG. 9A

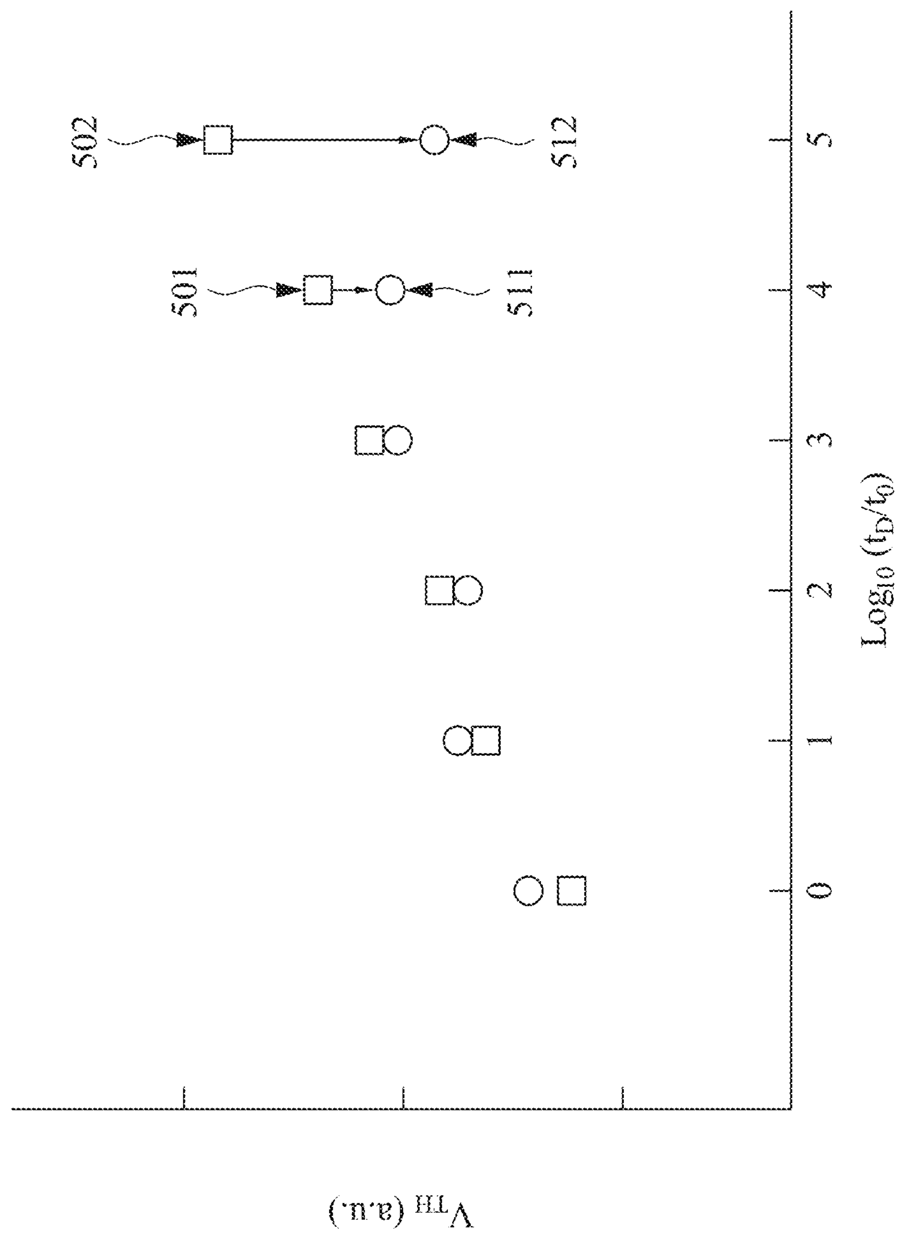
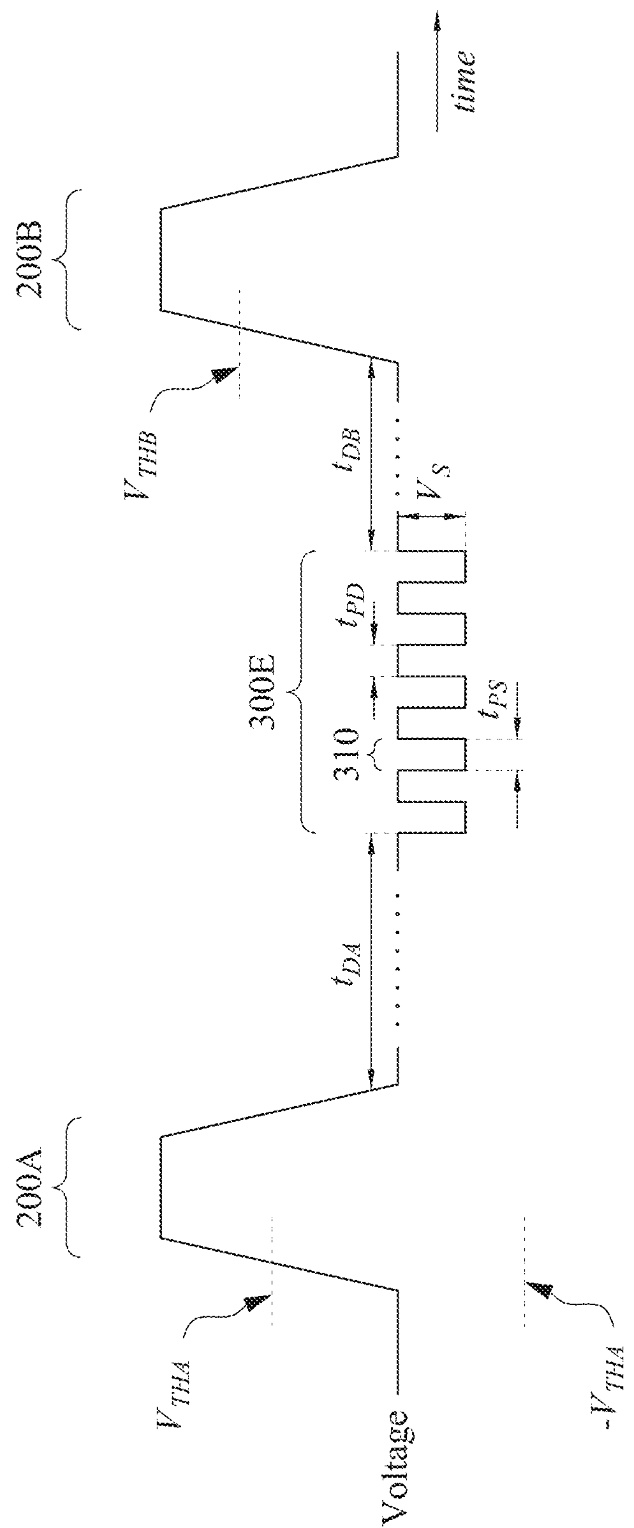


FIG. 9B



time
200B
200A
300E
310
V_{THB}
V_{THA}
$-V_{THA}$
t_{DB}
t_{DA}
t_{PD}
t_{PS}
V_S
Voltage

FIG. 10A

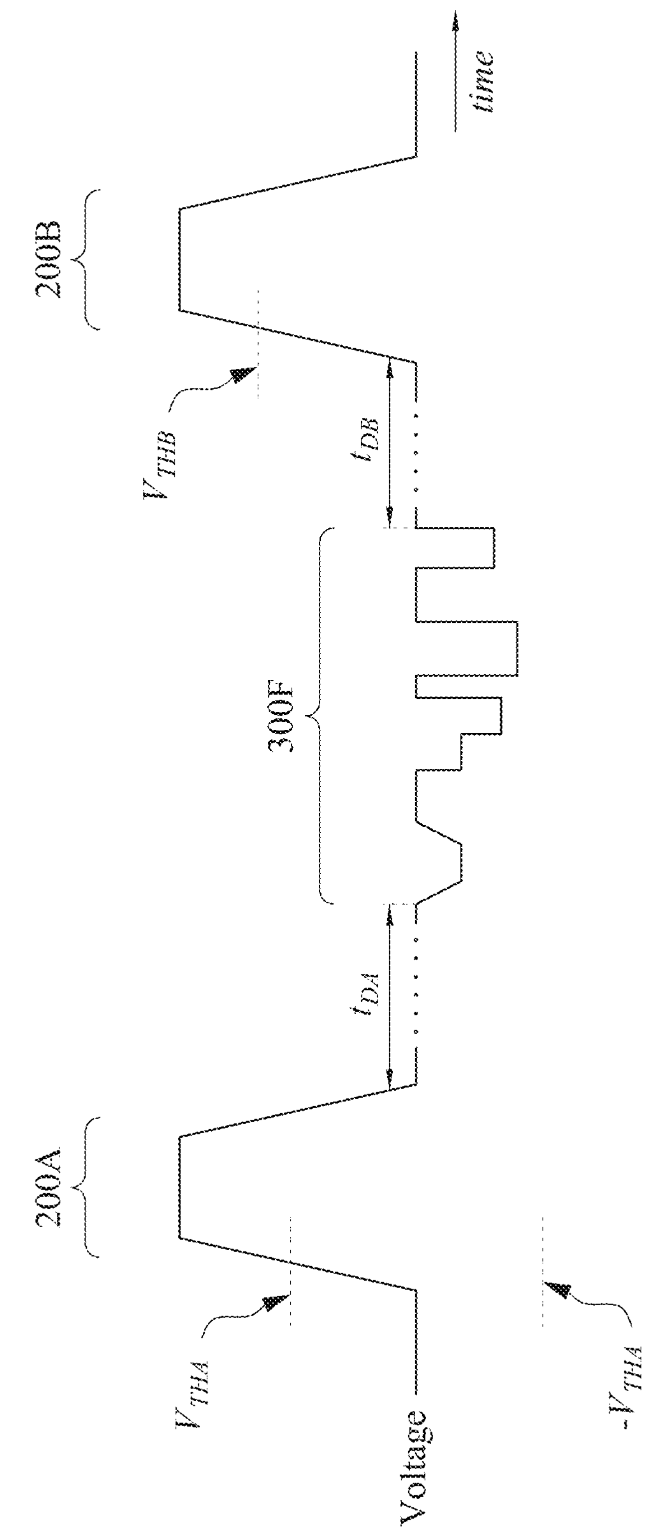


FIG. 10B

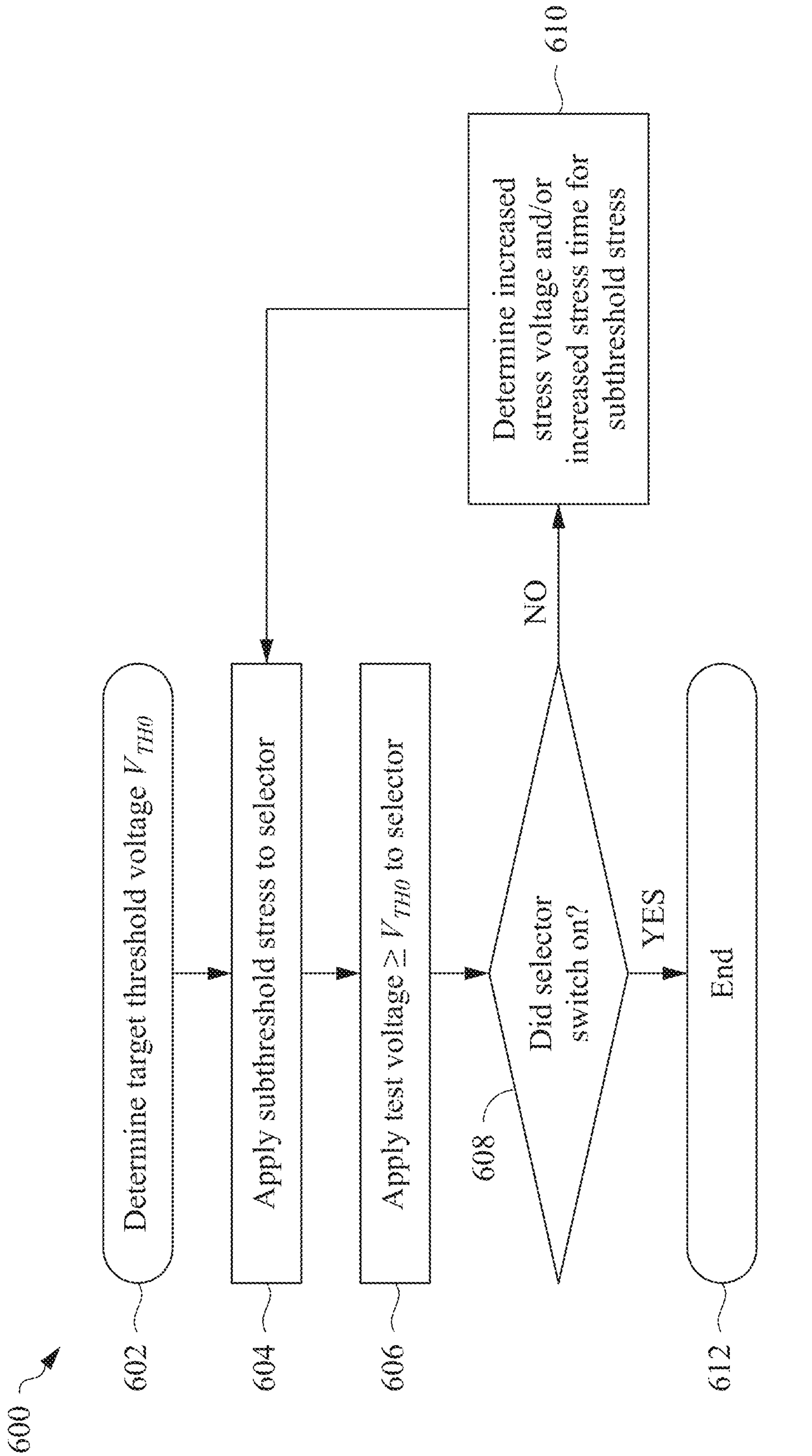

600
602
Determine target threshold voltage V_{TH0}
604
Apply subthreshold stress to selector
606
Apply test voltage ≥ V_{TH0} to selector
608
Did selector switch on?
YES
NO
610
Determine increased stress voltage and/or increased stress time for subthreshold stress
612
End

FIG. 11

MEMORY SELECTOR THRESHOLD VOLTAGE RECOVERY

PRIORITY CLAIM AND CROSS-REFERENCE

This application is a continuation of U.S. patent application Ser. No. 17/658,701, filed on Apr. 11, 2022, which claims the benefit of U.S. Provisional Application No. 63/267,621, filed on Feb. 7, 2022, each application is hereby incorporated herein by reference.

BACKGROUND

Semiconductor memories are used in integrated circuits for electronic applications, including radios, televisions, cell phones, and personal computing devices, as examples. One type of semiconductor memory is phase-change random access memory (PCRAM), which involves storing values in phase change materials, such as chalcogenide materials. Phase change materials can be switched between an amorphous phase (in which they have a high resistivity) and a crystalline phase (in which they have a low resistivity) to indicate bit codes. A PCRAM cell typically includes a phase change material (PCM) element between two electrodes. However, PCRAM cell selectors can be affected by a threshold voltage drift issue, in which the threshold voltage of the PCRAM cell drifts to higher values as a function of the elapsed time from a previous switching pulse (e.g., the delay time). For reliable device operation, it is beneficial to stabilize the threshold voltage over a broad range of delay times.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. **1** illustrates a perspective view of a memory array, in accordance with some embodiments.

FIGS. **2**A and **2**B illustrate cross-sectional views of memory cells, in accordance with some embodiments.

FIG. **3** illustrates a graph of the current-voltage characteristic of a selector, in accordance with some embodiments.

FIG. **4** illustrates a graph of the voltage and current across a selector, in accordance with some embodiments.

FIG. **5**A illustrates a graph of threshold voltage vs. delay time of a selector, in accordance with some embodiments.

FIG. **5**B illustrates a graph of the voltage and current across a selector, in accordance with some embodiments.

FIG. **6** illustrates a graph of the voltage across a selector including a subthreshold stress, in accordance with some embodiments.

FIG. **7** illustrates a graph of threshold voltage vs. delay time of a selector, in accordance with some embodiments.

FIGS. **8**A, **8**B, and **8**C illustrate graphs of the voltage across a selector including a subthreshold stress, in accordance with some embodiments.

FIG. **9**A illustrates a graph of the voltage across a selector including a continuous subthreshold stress, in accordance with some embodiments.

FIG. **9**B illustrates a graph of threshold voltage vs. delay time of a selector, in accordance with some embodiments.

FIGS. **10**A and **10**B illustrate graphs of the voltage across a selector including a subthreshold stress having multiple subthreshold voltage pulses, in accordance with some embodiments.

FIG. **11** illustrates a flowchart of an adaptive subthreshold stress process for a memory cell, in accordance with some embodiments.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as "beneath," "below," "lower," "above," "upper" and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

In accordance with some embodiments, a subthreshold voltage stress is applied to a threshold-type memory selector to counteract time delay threshold voltage drift. For example, the threshold voltage of a selector may increase during a time delay between switching pulses, and applying a subthreshold voltage stress as described herein can reduce the increased threshold voltage back toward a no-delay value. The subthreshold stress includes a voltage less than the threshold voltage of the selector, which can reduce the stressing of the selector, reduce power consumption, and reduce the chance of switching on the selector. The subthreshold voltage stress may also be a voltage with a polarity opposite to the switching voltage of the selector. By reducing threshold voltage using a subthreshold voltage stress as described herein, device uniformity, repeatability, or efficiency may be improved.

FIG. **1** illustrates a perspective view of a Phase-Change Random Access Memory (PCRAM) array **100**, in accordance with some embodiments. The PCRAM array **100** comprises word lines **102**, bit lines **104**, and PCRAM cells **110** connected in a "cross-point" configuration. Other configurations of word lines **102**, bit lines **104**, or PCRAM cells **110** are possible. In some embodiments, multiple PCRAM arrays **100** may be stacked to create a 3D memory array (not shown). The PCRAM array **100** may be formed on a substrate (not shown), which may be a semiconductor substrate or another type of substrate. The substrate may include active and/or passive devices (e.g., transistors, diodes, capacitors, resistors, or the like), in some embodiments. The devices may be formed according to applicable manufacturing processes. In some embodiments, no devices are formed in the substrate. In some embodiments, the PCRAM array **100** is formed in the metallization layers of an interconnect structure (not shown) over the substrate. The PCRAM array may be electrically connected to one or more of the metallization layers. For example, in some embodiments, the word lines **102** and/or the bit lines **104** may be conductive lines of the metallization layers.

In the embodiment shown in FIG. **1**, each PCRAM cell **110** comprises a top electrode **112**, a bottom electrode **114**, an ovonic threshold switching (OTS) layer **116**, and a phase-change material (PCM) layer **118**, described in greater detail below for FIGS. **2**A-**2**B. The bit lines **104** are electrically connected to the bottom electrodes **114** of respective columns of PCRAM cells **110** in the PCRAM array **100**. Each column of the PCRAM array **100** has an associated bit line **104**, and the PCRAM cells **110** in a column are connected to the bit line **104** for that column. The word lines **102** are connected to the top electrodes **112** of respective rows of PCRAM cells **110** in the PCRAM array **100**. Each row of the PCRAM array **100** has an associated word line **102**, and the PCRAM cells **110** in a row are connected to the word line **102** for that row. In this manner, each PCRAM cell **110** of the PCRAM array **100** may be selected by the appropriate combination of word line **102** and bit line **104**. For example, a particular PCRAM cell **110** may be selected (e.g., for reading or writing operations) by accessing the single word line **102** connected to that PCRAM cell **110** and also accessing the single bit line **104** connected to that PCRAM cell **110**.

In some embodiments, the resistance of the PCM layer **118** of each PCRAM cell **110** is programmable, and can be changed between a high-resistance state and a low-resistance state, which can correspond to the two states of a binary code. In some embodiments, the resistance state of the PCM layer **118** of a PCRAM cell **110** can be programmed (e.g., "written") by applying an appropriate electrical voltage pulse across the PCRAM cell **110** that generates a corresponding electrical current pulse across the PCM layer **118**. In some embodiments, the magnitude of a programming current pulse is in the range of about 50 μA to about 800μ, though other currents are possible. In some cases, the magnitude of a programming voltage pulse may be in the range of about 1 V to about 2 V, though other voltages are possible. In some embodiments, the state of a PCRAM cell **110** may be read by applying a relatively small electrical current across the PCRAM cell **110** that allows the resistance of the PCRAM cell **110** to be measured without disturbing the resistance state of the PCM layer **118**. Other types of memory or memory architecture may use different read schemes or magnitudes than this example.

The OTS layer **116** of each PCRAM cell **110** is used as a selector that allows the respective PCRAM cell **110** to be accessed (e.g., written or read) individually. In this manner, an OTS layer **116** of a PCRAM cell **110** may also be referred to herein as an "OTS selector **116**." An OTS selector **116** has a characteristic property called the threshold voltage (V_{TH}). At applied voltages below V_{TH} (e.g., at subthreshold voltages), the OTS selector **116** is in a high-resistance state, limiting current through the PCRAM cell **110**. At applied voltages greater than V_{TH}, the OTS selector **116** is in a low-resistance state that creates a current path through the PCRAM cell **110** for write or read operations. In this manner, write operations may be performed on a PCRAM cell **110** only when the voltage across the OTS selector **116** is greater than V_{TH}. In some cases, a PCRAM cell **110** may be read using subthreshold voltages. In some embodiments, the magnitude of the threshold voltage V_{TH} is in the range of about 1 V to about 2 V, though other voltages are possible. In some cases, the threshold voltage V_{TH} can be tuned, for example, by adjusting the materials or thicknesses of the various layers.

Turning briefly to FIG. **3**, an example graph of the current across an OTS selector **116** as a function of the voltage applied across the selector is shown. FIG. **3** is a representative graph for explanatory purposes, and OTS selectors **116** may have a different current-voltage characteristics than shown. As illustrated in FIG. **3**, an OTS selector **116** is in a low-conductivity state (e.g., an "off" state) unless a voltage above the on-state threshold voltage V_{TH} is applied, at which point the OTS selector **116** enters a high-conductivity state (e.g., an "on" state). The OTS selector **116** returns to the low-conductivity state when the applied voltage is removed or reduced below an off-state threshold voltage (also sometimes referred to as a "holding voltage"). The off-state threshold voltage may be the same as or different from the on-state threshold voltage V_{TH}. The threshold voltage V_{TH} of an OTS selector **116** of a PCRAM cell **110** may also be referred to herein as the threshold voltage V_{TH} of the PCRAM cell **110**.

Turning back to FIGS. **2**A and **2**B, cross-sectional views of PCRAM cells **110** are illustrated, in accordance with some embodiments. The PCRAM cells **110** in FIGS. **2**A and **2**B may be used in a PCRAM array such as the PCRAM array shown in FIG. **1**, or the like. The PCRAM cells **110** each comprise a top electrode **112**, a bottom electrode **114**, an OTS layer **116**, a PCM layer **118**, and an intermediate electrode **117**. One or more layers of insulating material **108** may surround and protect the PCRAM cells **110**, and may comprise one or more inter-metal dielectric (IMD) layers and/or one or more layers of oxide, nitride, or the like. FIG. **2**A illustrates a PCRAM cell **110** having a "symmetric" or "pillar" shape, and FIG. **2**B illustrates a PCRAM cell **110** having an "asymmetric" or "mushroom" shape with a narrower bottom electrode **114**. The PCRAM cells **110** shown in FIGS. **2**A and **2**B are non-limiting examples, and PCRAM cells **110** may have other layers, widths, configurations, dimensions, shapes, or configurations in other embodiments.

In some embodiments, the bit lines **104** may be formed in an insulating layer **106**, which may be an inter-metal dielectric (IMD) layer, or the like. The bit lines **104** may be formed of one or more conductive materials such as tungsten, titanium, tantalum, ruthenium, cobalt, nickel, the like, or combinations thereof, and may be deposited by a suitable process such as CVD, PVD, ALD, plating, or the like. The bit lines **104** may be formed using a suitable process, such as damascene, dual-damascene, or another process.

In some embodiments, the bottom electrodes **114**, the PCM layer **118**, the intermediate electrode **117**, the OTS layer **116**, and the top electrodes **112** are blanket deposited and then patterned together to form individual PCRAM cells **110**. The insulating material **108** may then be deposited around the PCRAM cells **110**. This process may form a pillar-shaped PCRAM cell **110** such as that shown in FIG. **2**A, for example. In other embodiments, one or more layers of the PCRAM cells **110** may be formed or patterned separately. This process may form a PCRAM cell **110** having layers of different widths, such as the mushroom-shaped PCRAM cell **110** shown in FIG. **2**B, in which the bottom electrodes **114** are formed and then the other layers of the PCRAM cell **110** are patterned in a separate step. For example, the bottom electrodes **114** may be formed first by etching a trench in a layer of insulating material **108** and filling the trench with conductive material. Other techniques are possible.

The bottom electrodes **114** are formed on the bit lines **104**. The bottom electrodes **114** may be formed from one or more conductive materials similar to those described for the bit lines **104**, and may be formed using similar processes. In some embodiments, the bottom electrodes **114** may include a barrier layer (not shown). The PCM layer **118** is formed on the bottom electrodes **114**. In some embodiments, the PCM layer **118** is formed of a chalcogenide material. Chalcogenide materials include at least a chalcogen anion (e.g., selenium (Se), tellurium (Te), or the like) and an electropositive element (e.g., germanium (Ge), silicon (Si), phosphorus (P), arsenic (As), antimony (Sb), bismuth (Bi), zinc (Zn), nitrogen (N), boron (B), carbon (C), or the like). An acceptable chalcogenide material includes, but is not limited to, GeSbTe (GST) or GeSbTeX, in which X is a material such as Ag, Sn, In, Si, N, or the like. Other materials are possible. The PCM layer **118** may be formed using a suitable deposition process, such as PVD, CVD, plasma-enhanced CVD (PECVD), ALD, or the like.

In some embodiments, an intermediate electrode **117** is formed on the PCM layer **118**. The intermediate electrode **117** may be formed using materials or techniques similar to those described for the bottom electrodes **114**. The OTS layer **116** is formed on the intermediate electrode **117**. The OTS layer **116** may be formed of one or more materials similar to those described above for the PCM layer **118**. For example, the OTS layer **116** may be formed of a chalcogenide material, which may be a similar material or a different material than the PCM layer **118**. The OTS layer **116** may be formed using a suitable deposition process, such as PVD, CVD, PECVD, ALD, or the like.

The top electrodes **112** are formed on the OTS layer **116**. The top electrodes **112** may be formed using materials or techniques similar to those described for the bottom electrodes **114**. The word lines **102** may then be formed on the top electrodes **112**. The word lines **102** may be formed using similar materials or techniques as the bit lines **104**, in some embodiments. Other materials or techniques are possible.

Turning to FIG. **4**, a representative graph of voltage and current of an OTS selector **116** (e.g., the OTS layer **116**) of a PCRAM cell **110** is shown, in accordance with some embodiments. The graph of FIG. **4** illustrates, as a function of time, a voltage applied across an OTS selector **116** and the corresponding current through the OTS selector **116**. FIG. **4** is a representative graph for explanatory purposes, and OTS selectors **116** may have a different current-voltage characteristics than shown. FIG. **4** illustrates a first voltage pulse **200**A followed by a second voltage pulse **200**B. Both pulses **200**A-B may be voltage pulses used for programming the PCRAM cell **110**, and thus both pulses **200**A-B include voltages greater than the threshold voltage V_{TH} of the OTS selector **116** of the PCRAM cell **110**. As shown in FIG. **4**, the second pulse **200**B is applied after a delay time t_D has elapsed from the completion of the first pulse **200**A. Depending on the particular application or operation of the PCRAM array **100**, a delay time t_D between two subsequent voltage pulses applied to a PCRAM cell **110** may encompass a wide range, such as a t_D as short as a fraction of a second (e.g., microseconds or less) or a t_D as long as many thousands of seconds (e.g., weeks or longer).

However, in some cases, the threshold voltage V_{TH} of an OTS selector **116** may increase as the time delay t_D increases. This "time delay threshold voltage drift" is shown in FIGS. **5**A and **5**B, which show a representative graph of V_{TH} vs. t_D and a representative graph of voltage and current across an OTS selector **116**, respectively. FIGS. **5**A-**5**B are representative graphs used for explanatory purposes, and OTS selectors **116** may have different characteristics than shown. FIG. **5**A illustrates an example relationship between the threshold voltage V_{TH} of an OTS selector **116** and the time delay t_D elapsed since the previous voltage pulse (e.g. pulse **200**A of FIG. **5**A). As shown in FIG. **5**A, the threshold voltage V_{TH} increases logarithmically with the time delay t_D. The increase of threshold voltage V_{TH} over time may be due to factors such as atomic relaxation of the material of the OTS layer **116**.

FIG. **5**B shows a graph of voltage and current of an OTS selector **116** illustrating an increase in threshold voltage after a time delay t_D. The graph of FIG. **5**B is similar to the graph of FIG. **4**, except that the first pulse **200**A has a threshold voltage V_{THA} and, after time delay t_D, the second pulse **200**B has an increased threshold voltage V_{THB}. The threshold voltage V_{THB} corresponds to the threshold voltage V_{TH} in FIG. **5**A. In some cases, the increase in the threshold voltage V_{TH} due to time delay t_D can reduce efficiency or necessitate the application of pulses of higher voltages, which can increase the risk of leakage, damage, or disturbing unselected PCRAM cells **110**. In some cases, the threshold voltage V_{TH} of a PCRAM cell **110** may increase to a voltage large enough that a writing and/or reading operation performed on that PCRAM cell **110** fails. In some cases, a "dummy" voltage pulse similar to the pulses **200**A-B may be applied between the first pulse **200**A and the second pulse **200**B to reset the increased threshold voltage (e.g., V_{THB}) to a lower voltage (e.g., V_{THA}) and effectively reset the time delay t_D to zero. However, the application of dummy voltage pulses in this manner can unnecessarily stress a PCRAM cell **110**, cause unwanted heating, and consume additional power.

FIG. **6** illustrates a representative graph of voltage applied across an OTS selector **116** of a PCRAM cell **110** including a subthreshold stress **300** between two voltage pulses **200**A-B, in accordance with some embodiments. Applying a subthreshold stress **300** as described herein can reduce the effects of time delay threshold voltage drift, described in greater detail below. The graph of FIG. **6** is similar to the graph of applied voltage in FIG. **5**B, except that a subthreshold stress **300** is applied to the OTS selector **116** at a time between the first pulse **200**A and the second pulse **200**B. In some embodiments, the subthreshold stress **300** comprises a pulse of voltage V_S ("the stress voltage") applied to the OTS selector **116** for a duration of time t_S ("the stress time"), in which the magnitude of the stress voltage V_S is less than the threshold voltage V_{THA} of the first pulse **200**A. The stress voltage V_S of the subthreshold stress **300** may have a polarity that is opposite the polarity of the pulses **200**A-B, in some embodiments. For example, the pulses **200**A-B may be a positive voltage and the stress voltage V_S may be a negative voltage, as shown in FIG. **6**. In this manner, the subthreshold stress **300** may be considered a "negative stress" or a "reverse polarity stress" in some cases. In other embodiments, the pulses **200**A-B and the stress voltage V_S may have the same polarity. In some cases, a subthreshold stress **300** comprising a V_S having a polarity opposite that of the pulses **200**A-B can improve the threshold voltage recovery effect of the subthreshold stress **300**, described in greater detail below.

In some embodiments, the subthreshold stress **300** comprises a voltage pulse applied to a PCRAM cell **110** for a stress time t_S that is in the range of about 1 millisecond to about 10 seconds, though other stress times are possible. The subthreshold stress **300** may comprise a stress voltage V_S that has a magnitude in the range of about 0.6 V to about 1.2 V, in some embodiments. Other stress voltages are possible, and a subthreshold stress **300** may comprise multiple different stress voltages V_S in other embodiments. In some embodiments, the magnitude of the stress voltage V_S may be based on or determined by the magnitude of a voltage threshold V_{TH} of the OTS selector **116** (e.g., V_{THA} or V_{THB}). For example, the stress voltage V_S may have a magnitude less than the magnitude of a voltage threshold V_{TH} of the OTS selector **116** (e.g., V_{THA} or V_{THB}). In some embodiments, the magnitude of V_S may be between about 50% and about 80% of the magnitude of V_{TH}. In some embodiments, the magnitude of V_S may be between about 200 mV and about 900 mV less than the magnitude of V_{TH}. Other relative magnitudes of V_S and V_{TH} are possible, and may be determined based on the magnitude of V_{TH}, for example. In some embodiments, the use of a subthreshold voltage pulse can reduce electrical stress on the PCRAM cell **110**, reduce power consumption, and avoid unwanted switching of other PCRAM cells **110**. The subthreshold stress **300** may comprise a substantially square or rectangular voltage waveform as shown in FIG. **6**, or the subthreshold stress **300** may have another waveform shape such as a triangular shape, trapezoidal shape, rounded shape, stepped shape, the like, or another shape. In some embodiments, the subthreshold stress **300** is applied at a first time delay t_{DA} after the first pulse **200**A, and the second pulse **200**B is applied at a second time delay t_{DB} after the subthreshold stress **300**. The time delays t_{DA} and t_{DB} may each be about zero microseconds or greater.

In some cases, the application of a subthreshold stress **300** as described herein can reduce the effects of time delay threshold voltage drift. This is illustrated in FIG. **7**, which shows measurement data of threshold voltage V_{TH} as a function of delay time t_D. The delay time t_D is shown in a logarithmic scale $\text{Log}_{10}(t_D/t_0)$, in which t_0 is a reference unit of delay time. The data points shown in FIG. **7** are intended as an illustrative example, and these or other threshold voltages may be present at these or other time delays in other embodiments. The square data points extending from data point **401** to data point **403** represent measurements of V_{TH} for an OTS selector **116** taken after various time delays t_D have elapsed. For example, the data point **403** represents a measured threshold voltage of V_{TH1} taken after a delay time of t_{D1}. During the time delay t_{D1}, the threshold voltage has increased from a baseline threshold voltage of approximately V_{TH0} to an "increased threshold voltage" of V_{TH1}. In this manner, the square data points of FIG. **7** indicate that V_{TH} increases with increasing t_D, similar to the relationship shown previously in FIG. **5**A.

The circle data point **410** represents a threshold voltage measurement taken for the same delay time t_{D1} as data point **403**, except the measurement of data point **410** was taken after first applying a subthreshold stress **300**. The applied subthreshold stress **300** was similar to the subthreshold stress **300** described for FIG. **6**. As shown in FIG. **7**, applying a subthreshold stress **300** has reduced the threshold voltage from an increased threshold voltage V_{TH1} to a "reduced threshold voltage" V_{TH2}. The magnitude of the threshold voltage reduction due to subthreshold stress **300** is indicated in FIG. **7** as V_{diff}.

In some embodiments, after the threshold voltage of an OTS selector **116** has increased to an increased threshold voltage (e.g., V_{TH1}) during a time delay, the increased threshold voltage may be decreased to a reduced threshold voltage (e.g., V_{TH2}) by application of subthreshold stress **300**. In other words, the utilization of a subthreshold stress **300** as described herein can lower the threshold voltage of an OTS selector **116** that has increased due to time delay. In this manner, applying subthreshold stress **300** can at least partially recover an increased threshold voltage towards a normal (e.g., a time delay of zero seconds) threshold voltage, and can thus reduce the effects of time delay threshold voltage drift. The data points shown in FIG. **7** are an example, and threshold voltage reduction using subthreshold stress **300** as described herein may be utilized for other time delays or threshold voltages than shown. The threshold voltage recovery benefits of subthreshold stress **300** as described herein are reproducible and repeatable, and can be implemented such that a PCRAM cell **110** experiences little or no permanent changes to operation or characteristics.

In some embodiments, the reduction of threshold voltage V_{diff} has a magnitude in the range of about 50 mV to about 500 mV. In some embodiments, the magnitude of V_{diff} may be between about 2% and about 40% of the magnitude of the increased threshold voltage. In some embodiments, the reduced threshold voltage may be between about 60% and about 98% of the magnitude of the increased threshold voltage. Other magnitudes or relative magnitudes are possible. The actual magnitude of threshold voltage reduction V_{diff} that is achieved may depend on the parameters of the subthreshold stress **300**, such as the first time delay t_{DA}, the second time delay t_{DB}, the stress voltage V_S, the stress time t_S, or other factors. For example, a second time delay t_{DB} that is greater than about zero seconds can allow the reduced voltage threshold to increase due to time delay threshold voltage drift. As another example, a larger V_S or a larger t_S may result in a larger V_{diff}. The threshold voltage reduction V_{diff} may be affected by other factors or conditions than these examples.

Still referring to FIG. **7**, data point **402** shows a measurement of increased threshold voltage approximately equal to V_{TH2}, which corresponds to a time delay of t_{D2}. In this manner, the threshold voltage reduction V_{diff} from V_{TH1} to V_{TH2} is approximately equivalent to reducing the actual time delay from t_{D1} to an effective time delay t_{D2}. The logarithmic difference between the actual time delay t_{D1} and the effective time delay t_{D2} is shown in FIG. **7** as effective delay reduction t_{diff}. For the data point **410**, the effective delay reduction t_{diff} corresponds to a reduction of about five orders of magnitude (e.g., t_{D2} is about $10^{-5} \times t_{D1}$). In other embodiments, the effective time delay reduction t_{diff} may be in the range of about one order of magnitude to about 12 orders of magnitude or larger, though other values are possible.

A subthreshold stress **300** may be controlled to have a variety of characteristics. The particular characteristics of a subthreshold stress **300** may be configured for a particular application or desired result, for example. FIGS. **8**A-**8**C, **9**A, and **10**A-**10**B illustrate graphs of voltages applied across OTS selectors **116** including subthreshold stresses **300**A-F, in accordance with some embodiments. The voltage graphs shown in FIGS. **8**A-**10**B are similar to the voltage graph shown in FIG. **6**, except for described differences in the characteristics of the subthreshold stresses **300**A-F. The subthreshold stresses **300**A-F are intended as non-limiting examples, and a subthreshold stress **300** may have different characteristics than shown or described while still remaining within the scope of the present disclosure.

The stress voltage or stress time of a subthreshold stress **300** may be controlled to provide a suitable reduction of threshold voltage. For example, FIG. **8**A illustrates a subthreshold stress **300**A having a relatively small stress voltage V_{SA} and a relatively long stress time t_{SA}. In some cases, applying a smaller magnitude of stress voltage for a longer stress time can reduce electrical stress or heating of the PCRAM cell **110**. FIG. **8**B illustrates a subthreshold stress **300**B having a relatively large stress voltage V_{SB} and a relatively short stress time t_{SB}. In some embodiments, a larger stress voltage or longer stress time may achieve a larger threshold voltage reduction V_{diff}. In some embodiments, the stress voltage or stress time may be controlled to depend on the first time delay t_{DA}. For example, after a longer first time delay t_{DA}, a subthreshold stress **300** may be controlled to have a larger stress voltage or longer stress time to achieve a larger threshold voltage reduction V_{diff}. In some embodiments, a similar threshold voltage reduction V_{diff} may be achieved by applying a subthreshold stress **300** having a relatively small stress voltage V_S and a relatively long stress time t_S (e.g., similar to subthreshold stress **300**A) or by applying a subthreshold stress **300** having a relatively large stress voltage V_S and a relatively short stress time t_S (e.g., similar to subthreshold stress **300**B).

In some embodiments, a subthreshold stress **300** comprises multiple stress voltages that each may have a corresponding stress time. FIG. **8**C illustrates a continuous subthreshold stress **300**C that includes a first stress voltage V_{SC1} followed by a second stress voltage V_{SC2} that is different from V_{SC1}. The first stress voltage V_{SC1} is applied for a stress time t_{C1} and the second stress voltage V_{SC2} is applied for a stress time t_{C2} that is different from t_{C1}. A subthreshold stress **300** may comprise more than two different stress voltages in other embodiments. The various stress voltages may be sequentially arranged in any suitable order. For example, in some embodiments, the second stress voltage V_{SC2} may be applied before the first stress voltage V_{SC1}. Any suitable sequential arrangement of multiple stress voltages within a subthreshold stress **300** may be used. The various stress voltages may have the same stress times or may have different stress times. For example, in some embodiments, the first stress voltage V_{SC1} may be applied for t_{C2} and the second stress voltage V_{SC2} may be applied for t_{C1}. Any suitable stress times may be used for the various stress voltages within a subthreshold stress **300**.

FIG. **9**A illustrates a subthreshold stress **300**D that is applied continuously between the first voltage pulse **200**A and the second voltage pulse **200**B, in accordance with some embodiments. In other words, the stress time t_S of the subthreshold stress **300**D is equal to the delay time t_D between the pulses **200**A-B. As shown in FIG. **9**A, the subthreshold stress **300**D may have a stress voltage polarity that is opposite from the polarity of the voltage pulses **200**A-B, in some embodiments. The subthreshold stress **300**D shown in FIG. **9**A has a constant V_S, but in other embodiments the stress voltage V_S may vary during application of a continuous subthreshold stress **300**D. In some cases, applying a continuous subthreshold stress **300**D may allow the effects of time delay threshold voltage drift to be reduced while allowing for little or no monitoring or measuring of the delay time t_D. Additionally, the use of a stress voltage V_S that is a subthreshold voltage can reduce heating or power consumption.

FIG. **9**B shows measurement data of threshold voltage V_{TH} as a function of delay time t_D with and without an applied continuous subthreshold stress **300**D. The delay time t_D is shown in a logarithmic scale $\mathrm{Log}_{10}(t_D/t_0)$, in which t_0 is a reference unit of delay time. The circular data points represent measurements of threshold voltage V_{TH} for an OTS selector **116** taken after various time delays t_D have elapsed, with a subthreshold stress **300**D continuously applied during the time delay t_D. The square data points represent measurements of threshold voltage V_{TH} for an OTS selector **116** taken after various time delays t_D have elapsed, but without any subthreshold stress applied. FIG. **9**B shows that the application of continuous subthreshold stress **300**D reduces the effects of time delay threshold voltage drift. For example, the application of continuous subthreshold stress **300**D reduces the increased threshold voltage of data point **501** to the reduced threshold voltage of data point **511**, and the application of continuous subthreshold stress **300**D reduces the increased threshold voltage of data point **502** to the reduced threshold voltage of data point **512**. In some cases, the application of a continuous subthreshold stress **300**D can result in a greater reduction of increased threshold voltage for longer delay times t_D. The measurement data shown in FIG. **9**B is a non-limiting example shown for explanatory purposes.

In some embodiments, a single subthreshold stress **300** may comprise two or more subthreshold voltage pulses **310**. This is shown in FIG. **10**A, in which the subthreshold stress **300**E is formed from a plurality of subthreshold voltage pulses **310** of stress voltage V_S. The subthreshold voltage pulses **310** of the subthreshold stress **300**E each have a pulse stress time t_{PS}, and the subthreshold voltage pulses **310** are separated by a pulse delay time t_{PD}. The subthreshold stress **300**E may have fewer or more subthreshold voltage pulses **310** than shown in FIG. **10**A. In some cases, the utilization of a subthreshold stress **300**E comprising multiple subthreshold voltage pulses **310** can avoid effects due to heating accumulation or electric field stress, particularly for large stress voltages V_S. Various characteristics of the subthreshold stress **300**E may be controlled, such as the period of the subthreshold voltage pulses **310** (e.g., $t_{PS}+t_{PD}$), the duty cycle of the subthreshold voltage pulses **310** (e.g., $t_{PS}:t_{PD}$), the stress voltage V_S, the number of subthreshold voltage pulses **310**, or other characteristics.

In other embodiments, a single subthreshold stress **300** may comprise multiple subthreshold voltage pulses having different stress voltages V_S, different pulse stress times t_{PS}, or different pulse delay times t_{PD}. A non-limiting example is shown in FIG. **10**B, in which a single subthreshold stress **300**F comprises multiple subthreshold voltage pulses having various different characteristics such as pulse stress time, pulse delay time, stress voltage, waveform, etc. All variations and combinations of subthreshold voltage pulses are considered within the scope of the present disclosure.

A subthreshold stress **300** as described herein may be applied to a single PCRAM cell **110** or applied in parallel to multiple PCRAM cells **110**. For example, a subthreshold stress **300** may be simultaneously applied to PCRAM cells **110** that share the same word line **102** or the same bit line **104**. In some cases, the use of a subthreshold stress voltage V_S as described herein can reduce power consumption when applying a subthreshold stress **300** to multiple PCRAM cells **110**. A subthreshold stress **300** as described herein may be applied in various ways, which may depend, for example, on a particular application, device, or device operation. For example, in some embodiments, a subthreshold stress **300** may be applied after a device (or portion of a device) is powered on or activated. In some embodiments, a subthreshold stress **300** may be if it is determined that the delay time t_D is at or above a certain threshold delay time (e.g., a certain number of seconds or the like). In some embodiments, a subthreshold stress **300** may be applied periodically. In some embodiments, a subthreshold stress **300** may not be applied if the delay time t_D is less than a certain threshold delay time. In some embodiments, the characteristics of the subthreshold stress **300** may be controlled based on the delay time t_D.

For example, a smaller t_D may correspond to a smaller V_S or a smaller number of subthreshold voltage pulses. Other examples are possible. In some embodiments, a subthreshold stress **300** may be applied if it is determined that the threshold voltage V_{TH} has increased by a certain amount of voltage or has increased to a voltage that is at or above a certain threshold voltage. Other conditions or techniques for applying or controlling a subthreshold stress **300** are possible.

In some embodiments, a subthreshold stress **300** may be applied adaptively to reduce an increased threshold voltage of a PCRAM cell **110** to a threshold voltage near or below a target threshold voltage V_{TH0}. This can reduce variation and put a PCRAM cell **110** in a more definite or desirable condition. In some embodiments, threshold voltage variation among multiple PCRAM cells **110** can be reduced by separately applying adaptive subthreshold stresses **300** to each of the PCRAM cells **110**. In this manner, the multiple PCRAM cells **110** can all simultaneously have a threshold voltage near or below a single target threshold voltage V_{TH0}. This can improve the uniformity of PCRAM cells **110** in a PCRAM array **100**, for example. In some cases, using an adaptive subthreshold stress process can increase the effectiveness of applying a subthreshold stress **300**, optimize the stress applied to an OTS selector, or avoid over-stressing the OTS selector (e.g., stressing more than necessary). In some embodiments, a subthreshold stress **300** may be adaptively applied by repeatedly first measuring the threshold voltage of the PCRAM cell **110** and then applying a subthreshold stress **300**, in which the characteristics of each applied subthreshold stress **300** are based on a difference between the target threshold voltage V_{TH0} and the previously measured threshold voltage. For example, the stress voltage V_S or the stress time t_S of an applied subthreshold stress **300** may be relatively smaller when a previously measured threshold voltage is relatively closer to the target threshold voltage V_{TH0}. Other techniques are possible.

FIG. **11** illustrates a flowchart of an adaptive subthreshold stress process **600** for a PCRAM cell, in accordance with some embodiments. At step **602**, a target threshold voltage V_{TH0} for the PCRAM cell is determined. The adaptive subthreshold stress process **600** may be performed, for example, to reduce the threshold voltage of a PCRAM cell below the target threshold voltage V_{TH0} or to adjust the threshold voltage of a PCRAM cell to be approximately equal to the target threshold voltage V_{TH0}. In some cases, the adaptive subthreshold stress process **600** may be performed with the assumption that time delay threshold voltage drift has increased the threshold voltage of the PCRAM cell above the target threshold voltage V_{TH0}. At step **602**, an initial subthreshold stress is applied to the PCRAM cell. The initial subthreshold stress may be similar to any of the subthreshold stresses described herein. For example, the initial subthreshold stress may comprise one subthreshold voltage pulse or multiple subthreshold voltage pulses, may have a stress voltage V_S and a stress time t_S, etc. The initial subthreshold stress can reduce the threshold voltage of the PCRAM cell toward the target threshold voltage V_{TH0}.

At step **606**, a test voltage of at or above the target threshold voltage V_{TH0} is applied to the PCRAM cell. A test voltage approximately equal to the target threshold voltage V_{TH0} may be applied if reducing the threshold voltage below the target threshold voltage V_{TH0} is desired. A test voltage approximately equal to the target threshold voltage V_{TH0} or slightly larger than the target threshold voltage V_{TH0} may be applied if adjusting the threshold voltage to be approximately equal to the target threshold voltage V_{TH0} is desired. If the initial subthreshold stress has sufficiently reduced the threshold voltage of the PCRAM cell, the test voltage applied at step **606** switches on the OTS selector of the PCRAM cell, bringing the OTS selector temporarily into a high-conductivity state (e.g., the "on" state).

At step **608**, whether or not the OTS selector was switched on by the applied test voltage at step **606** is measured or otherwise detected. If the OTS selector was successfully switched on at step **606**, then the threshold voltage of the PCRAM cell has been reduced to approximately the target threshold voltage V_{TH0} or below the target threshold voltage V_{TH0}. In this case of successful switching, the process **600** ends at step **612**.

If the threshold voltage of the PCRAM cell is still too high, then the voltage applied at step **606** is not sufficient to switch on the OTS selector, and the process **600** continues from step **608** to step **610**. At step **610**, new subthreshold stress characteristics are determined that may provide a larger threshold voltage reduction effect than the initial subthreshold stress. For example, in some embodiments, a new stress voltage may be determined that is larger than the stress voltage of the initial subthreshold stress, and/or a new stress time may be determined that is longer than the stress time of the initial subthreshold stress. In some embodiments, the number of subthreshold voltage pulses may be increased from the number used in the initial subthreshold stress. Other characteristic changes to determine a new subthreshold stress characteristics are possible.

After determining the new subthreshold stress characteristics in step **610**, the process **600** returns to step **604** and applies a subthreshold stress with the new subthreshold stress characteristics. In some cases, new subthreshold stress characteristics are not determined at step **610**, and the subthreshold stress at step **604** re-uses previously determined subthreshold stress characteristics. In other embodiments, and the subthreshold stress at step **604** always re-uses the same subthreshold stress characteristics. The process **600** then applies a test voltage at step **606** and determines if the OTS selector was switched on at step **608**, as before. If the OTS selector was switched on, the process **600** ends at step **612**. If the OTS selector was not switched on, new subthreshold stress characteristics are determined in step **610**, which may again include increasing the stress voltage, stress time, and/or number of subthreshold voltage pulses. A subthreshold stress with the new subthreshold stress characteristics is applied at step **604**, and in this manner the process **600** may continue determining new subthreshold stress characteristics until the OTS selector is successfully switched on at step **608** and the process **600** ends at step **612**.

The use of a subthreshold stress to reduce the effects of time delay threshold voltage drift of a memory cell selector as described herein has advantages. The techniques described herein allow for at least a partial recovery of an increased threshold voltage of a memory cell selector toward a baseline threshold voltage, while keeping the memory cell selector in an off-state. The techniques described herein may be applied to a single selector or multiple selectors in parallel, for example, by applying the subthreshold stress to the selectors sharing the same word line or bit line. The subthreshold stress uses voltages below the voltage threshold rather than a voltage at or above the voltage threshold. This can avoid the necessity of a large voltage for threshold voltage recovery after a long delay time. The use of subthreshold voltages can reduce power consumption, reduce electric field or heating stress of the selector, or reduce the risk of switching on neighboring selectors. The subthreshold stress may have a voltage that is opposite in polarity to the selector switching voltage. The characteristics of the subthreshold stress may be configured, for example, to optimize threshold voltage recovery, improve efficiency, or minimize power consumption. In some cases, the characteristics of the subthreshold stress can be adaptively configured to provide a desired threshold voltage recovery. The techniques described herein are described for PCRAM, but may be applied to other types of non-volatile memory such as RRAM, MRAM, SOT-MRAM, or the like. The techniques described herein may be applied to a variety of memory architectures, such as 1-selector-1-resistor (1S1R) configurations, cross-point or crossbar configurations, stacked configurations of multiple layers, 3D vertical configurations, or the like. Additionally, the techniques described herein may be applied to memories comprising OTS selectors, chalcogenide selectors, or other threshold-type selectors or selector materials that exhibit time delay threshold memory drift.

In accordance with some embodiments of the present disclosure, a method includes applying a first voltage pulse across a memory cell, wherein the memory cell includes a selector, wherein the first voltage pulse switches the selector into an on-state; after applying the first voltage pulse, applying a second voltage pulse across the memory cell, wherein before applying the second voltage pulse the selector has a first voltage threshold, wherein after applying the second voltage pulse the selector has a second voltage threshold that is less than the first voltage threshold; and after applying the second voltage pulse, applying a third voltage pulse across the memory cell, wherein the third voltage pulse switches the selector into an on-state; wherein the selector remains continuously in an off-state between the first voltage pulse and the third voltage pulse. In an embodiment, the first voltage pulse and the third voltage pulse have a first voltage polarity, and the second voltage pulse has a second voltage polarity that is opposite from the first voltage polarity. In an embodiment, the second voltage pulse has a magnitude that is less than the magnitude of the first voltage threshold and less than the magnitude of the second voltage threshold. In an embodiment, the method includes applying a fourth voltage pulse after applying the first voltage pulse and before applying the second voltage pulse. In an embodiment, the selector includes a layer of GeCTe. In an embodiment, a duration of time between the first voltage pulse and the second voltage pulse is greater than one second. In an embodiment, the third voltage pulse is applied immediately after the second voltage pulse is applied. In an embodiment, the first voltage threshold is greater than the second voltage threshold by a voltage difference in the range of 50 mV to 500 mV. In an embodiment, the second voltage pulse is applied continuously from the first voltage pulse to the third voltage pulse.

In accordance with some embodiments of the present disclosure, a method includes performing a subthreshold stress process on a memory array, wherein each memory cell of the memory array includes a chalcogenide selector, wherein performing the subthreshold stress process includes applying a first voltage between a first word line and a first bit line of the memory array, wherein a first memory cell of the memory array is connected to the first word line and the first bit line, wherein at the start of the subthreshold stress process the chalcogenide selector of the first memory cell has a first threshold voltage, wherein a magnitude of the first voltage is less than the first threshold voltage, wherein at the end of the subthreshold stress process the chalcogenide selector of the first memory cell has a second threshold voltage that is smaller than the first threshold voltage. In an embodiment, performing the subthreshold stress process includes simultaneously applying the first voltage between the first word line and a second bit line of the memory array, wherein a second memory cell of the memory array is connected to the first word line and the second bit line. In an embodiment, applying the first voltage includes applying a series of pulses of the first voltage. In an embodiment, the method includes, before performing the subthreshold stress process, applying a second voltage between the first word line and the first bit line of the memory array, wherein the second voltage is greater than the first threshold voltage, wherein a polarity of the second voltage is opposite that of the first voltage, wherein no voltage is applied between the first word line and the first bit line in the time between the applying of the second voltage and the applying of the first voltage. In an embodiment, the method includes determining an elapsed time since the applying of the second voltage, wherein the applying of the first voltage is based on the elapsed time. In an embodiment, the memory array is a Phase-Change Random Access Memory (PCRAM) array.

In accordance with some embodiments of the present disclosure, a method includes determining a target threshold voltage for a chalcogenide selector; applying a first voltage pulse to the chalcogenide selector, wherein a magnitude of the first voltage pulse is less than the target threshold voltage; after applying the first voltage pulse, applying a test voltage pulse to the chalcogenide selector, wherein the magnitude of the test voltage pulse is at least the magnitude of the target threshold voltage; determining whether the test voltage pulse successfully switched the chalcogenide selector into an on-state; and based on the determining of the switching of the chalcogenide selector into an on-state, applying a second voltage pulse to the chalcogenide selector, wherein a duration of the second voltage pulse is longer than a duration of the first voltage pulse and/or a magnitude of the second voltage pulse is larger than a magnitude of the first voltage pulse. In an embodiment the method includes, based on the determining of the switching of the chalcogenide selector into an on-state, applying a third voltage pulse to the chalcogenide selector before applying the second voltage pulse. In an embodiment, the magnitude of the second voltage pulse is less than the target threshold voltage. In an embodiment, the chalcogenide selector is the selector of a memory cell of a PCRAM memory array. In an embodiment, the second voltage pulse is not applied after determining that the chalcogenide selector was successfully switched into an on-state.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A method comprising:

applying a first voltage between a first word line and a first bit line wherein a magnitude of the first voltage is greater than a first threshold voltage of a first selector that is electrically coupled to the first word line and the first bit line;

after applying the first voltage, applying no voltage between the first word line and the first bit line for a first nonzero duration of time; and

after the first nonzero duration of time, applying a second voltage between the first word line and the first bit line, wherein a magnitude of the second voltage is less than the first threshold voltage, wherein the first voltage and the second voltage have opposite polarities, wherein after applying the second voltage, the first selector has a second threshold voltage that is smaller than the first threshold voltage.

2. The method of claim **1** further comprising, after applying the second voltage, applying a third voltage between the first word line and the first bit line, wherein the third voltage and the second voltage have a same polarity.

3. The method of claim **2**, wherein between applying the second voltage and applying the third voltage, no voltage is applied between the first word line and the first bit line for a second nonzero duration of time.

4. The method of claim **2**, wherein a duration of the second voltage and a duration of the third voltage are different.

5. The method of claim **2**, wherein the magnitude of the second voltage and a magnitude of the third voltage are different.

6. The method of claim **1**, wherein a duration of the second voltage is greater than a duration of the first voltage.

7. The method of claim **1**, wherein the first threshold voltage is greater than the second threshold voltage by a voltage difference in the range of 50 mV to 500 mV.

8. The method of claim **1**, wherein the magnitude of the second voltage is based on a duration of the first nonzero duration of time.

9. The method of claim **1**, wherein the magnitude of the second voltage is based on the magnitude of the first threshold voltage.

10. A method comprising:

applying a first positive voltage pulse across a chalcogenide selector, wherein the first positive voltage pulse turns on the chalcogenide selector;

after applying the first positive voltage pulse, turning off the chalcogenide selector for at least one microsecond;

applying a first negative voltage pulse across the chalcogenide selector, wherein the first negative voltage pulse does not turn on the chalcogenide selector; and

after applying the first negative voltage pulse, applying a second positive voltage pulse across the chalcogenide selector, wherein the second positive voltage pulse turns on the chalcogenide selector.

11. The method of claim **10** further comprising, between applying the first negative voltage pulse and the second positive voltage pulse, applying a second negative voltage pulse across the chalcogenide selector.

12. The method of claim **10** further comprising, between applying the first negative voltage pulse and the second positive voltage pulse, turning off the chalcogenide selector for at least one microsecond.

13. The method of claim **10**, wherein applying the second positive voltage pulse comprises performing a reading operation on a memory cell.

14. The method of claim **10**, wherein applying the first negative voltage pulse reduces a threshold voltage of the chalcogenide selector.

15. The method of claim **10**, wherein a duration of the first negative voltage pulse is shorter than a duration of the first positive voltage pulse.

16. The method of claim **10**, wherein the first positive voltage pulse is a writing operation for a memory cell.

17. A method comprising:

performing a first writing operation on a memory cell, wherein the memory cell comprises an ovonic threshold switching (OTS) layer, wherein the first writing operation comprises a first voltage; and

at least one microsecond after performing the first writing operation, performing a subthreshold stress operation on the memory cell, wherein the subthreshold stress operation comprises a second voltage that is opposite in polarity to the first voltage.

18. The method of claim **17**, wherein the first voltage is greater than a threshold voltage of the OTS layer and the second voltage is less than the threshold voltage of the OTS layer.

19. The method of claim **17**, wherein the OTS layer comprises GeCTe.

20. The method of claim **17** further comprising, after performing the subthreshold stress operation, performing a second writing operation on the memory cell.

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